4M17 Coursework Assignment 1

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Coursework Assignment Overview

In this assignment, you will develop an optimiser for convex optimisation problems and use it to train a data classification model. You are required to produce a **technical report** detailing your approach, results, and analysis.

Your report should clearly explain the key steps you have taken to complete the coursework. Simply stating answers or presenting results without explanation will receive few marks. Figures and plots should be carefully designed and presented, and should always be accompanied by clear and concise explanations.

You can include *mathematical* assertions and error handling in your code, such as ensuring that a problem is feasible, but do not spend too much time validating inputs. It can be assumed that, for example, matrices have compatible dimensions, or that the correct number of gradient and Hessian functions have been provided.

Please carefully follow the submission and formatting guidelines below:

- The deadline for submitting your report is given on the course Moodle page. You must submit an electronic copy via Moodle in PDF format, including a completed 2026/27 coursework cover sheet. *You should include only your Coursework Candidate Number (CCN), and must ensure that your name does* ***not*** *appear anywhere in the report or in any file names.*
- *Marks will be deducted if you fail to use a coursework cover sheet.*
- Attach the source code for all programs you have developed as appendices to your report.
- The report should not exceed **10 pages** in length, excluding source code appendices.
- This assignment contributes **50% of your final course grade**. Questions related to the assignment should be directed to **Alex Grafton (ajg206)**.
- The **approximate** percentage of the assignment's marks allocated to each part of a question is indicated in square brackets.
- You are reminded of the Department of Engineering's *Use of Generative AI Policy (June 2026)*[1], and the need to complete the appropriate field in the cover sheet, even if no AI has been used. Some paraphrased sections of the policy are as follows; this list is not exhaustive:
  - Producing drafts of sections, paragraphs and illustrative content is not permitted.
  - Supporting refinement of style, clarity and structure, and highlighting grammatical issues is permitted in this work. Occasional minor spelling or grammatical errors that do not affect understanding will not be penalised.
  - Assisting in typesetting and formatting is permitted, provided you are using AI to help implement your intended layout, and not to design the layout itself.
  - Using AI for code generation is permitted, but you must understand the logic of the code before use. Tests and checks to verify your solver should be your own work.
  - It is advised that mathematical derivations be your own work, as the text accompanying them *must* be your own work.

[1] https://teaching.eng.cam.ac.uk/node/7600

Questions

1. In this question, you will develop a solver for convex optimisation problems. You may use any programming language you wish (Python is recommended), but *do not* call off-the-shelf optimisation routines – use only basic linear-algebra operations (from e.g. NumPy).

   In your report, you should describe your implementation and any measures you have taken to ensure it is correct. You may wish to include key code snippets to aid your explanation.

   Unless otherwise stated, use backtracking line search with parameters $\alpha = 0.1$ and $\beta = 0.5$, and terminate the Newton solver when the Newton decrement λ satisfies $\lambda(x)^2/2 \leq 10^{-6}$.

   (a) Consider the convex function

   $$f_0(x) = \log\left(e^{\frac{1}{2}x_1} + e^{x_2}\right) + \tfrac{1}{2}\|x\|_2^2,$$

   where $x \in \mathbb{R}^2$ and log is the natural logarithm. Calculate the gradient and Hessian matrix of f_0.

   [3%]

   (b) Using the Newton solver implementation provided on Moodle, solve the unconstrained optimisation problem

   $$\text{Minimise} \quad f_0(x),$$

   where the expression for f_0 is given in part (a). Use the starting point $x_0 \coloneqq (-3/2, 1)$. In your report, give the computed optimal solution and plot the iterate sequence of xs over level sets of f_0.

   [7%]

   (c) Extend the Newton solver from Moodle to solve the equality-constrained problem

   $$\begin{aligned} \text{Minimise} \quad & f_0(x) \\ \text{subject to} \quad & Ax = b, \end{aligned}$$

   where f_0 is a twice-differentiable, convex function, $A \in \mathbb{R}^{p\times n}$, and $b \in \mathbb{R}^p$. Assume that $\operatorname{rank}(A) = p < n$. Use your solver to solve the problem with $f_0(x)$ defined as in part (a), $A = \begin{bmatrix} 1.5 & 1 \end{bmatrix}$, and $b = \begin{bmatrix} 0.5 \end{bmatrix}$. As a starting point x_0, take the solution to the constraint equations with the smallest ℓ_2 norm. State the computed optimal solution in your report.

   [7%]

   (d) Consider the following convex optimisation problem

   $$\begin{aligned} \text{Minimise} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, \quad i = 1, \ldots, m, \\ & Ax = b, \end{aligned}$$

   where $f_0, \ldots, f_m$ are twice-differentiable, convex functions, $A \in \mathbb{R}^{p\times n}$, and $b \in \mathbb{R}^p$. Take f_0 as in part (a), A and b as in part (c), and the following inequality constraint functions:

   $$\begin{aligned} f_1(x) &= x_1 + x_2 - \frac{3}{2}, \\ f_2(x) &= -\frac{1}{4}x_1 + x_2 - 2, \\ f_3(x) &= \frac{1}{3}x_1^2 + \frac{5}{6}x_1 - x_2 + 1. \end{aligned}$$

   Describe the geometry of the feasible set.

   [3%]

Question 1 continues on the following page

(e) Implement the barrier method for the problem in part (d), using your equality-constrained Newton solver for the inner solves. Stop the outer loop when $m/t \leq 10^{-6}$, where t is the logarithmic barrier parameter. To obtain a starting point x_0, you should use your barrier method implementation to solve the phase I problem

$$\begin{aligned} \text{Minimise} \quad & s \\ \text{subject to} \quad & f_i(x) \leq s, \quad i = 1, \ldots, m, \\ & Ax = b. \end{aligned}$$

Report the computed solutions to the phase I and optimisation problems.

[10%]

(f) Consider the following optimisation problem

$$\begin{aligned} \text{Minimise} \quad & f_0(x) \\ \text{subject to} \quad & f_i(x) \leq 0, \quad i = 1, \ldots, m, \end{aligned}$$

where

$$\begin{aligned} f_1(x) &= x_1 + x_2 - \frac{3}{2}, \\ f_2(x) &= -\frac{1}{4}x_1 + x_2 - 2, \\ f_3(x) &= \sin(4x_1) + \frac{1}{3}x_1^2 + \frac{5}{6}x_1 - x_2 + 1. \end{aligned}$$

Take f_0 as in part (a). Note that this is the same as the problem in part (d) without the equality constraint and with a different f_3, which is no longer convex. Describe the geometry of the feasible set, and explore what happens when you solve the phase I and optimisation problems. Explain why this method may not succeed. *(You do not need to precisely derive the conditions to arrive at the global optimum.)*

[7%]

(g) Include the equality constraint from part (c) in the problem in part (f). Describe the geometry of the feasible set. By considering the feasible set, explain how the inequality constraints or solution method could be modified to ensure the global optimum can be found reliably. *(An explanation will suffice – you do not need to solve the new problem.)*

[3%]

(h) Consider the following phase I problem with affine inequality constraints and no equality constraints:

$$\begin{aligned} \text{Minimise} \quad & s \\ \text{subject to} \quad & Gx - h \preceq s\mathbf{1}, \end{aligned}$$

where $G \in \mathbb{R}^{m \times n}$, $h \in \mathbb{R}^m$, and $\mathbf{1} \in \mathbb{R}^m$ is the vector with all entries equal to 1. When solving this with the barrier method, under what conditions on G is the Newton step well defined?

[7%]

(i) Briefly discuss the main performance bottleneck in your solver. Suggest one or two approaches to improve scalability for large problems.

[3%]

2. Suppose we are given a set of training data points $x_i \in \mathbb{R}^n$, $i = 1, \ldots, m$, each associated with a label $y_i \in \{-1, 1\}$. For example, each x_i might represent the results of a particular medical test, with $y_i = 1$ indicating the presence of a disease and $y_i = -1$ its absence.

Our goal is to construct a **classifier** that predicts the label of a new data point x_{new} based on the training data. We assume that the data are **linearly separable** – that is, there exists a hyperplane

$$w^T x + b = 0, \quad w \in \mathbb{R}^n,\ b \in \mathbb{R},$$

which perfectly separates the positive and negative examples. For a new point x_{new}, we classify it as

$$\hat{y}_{\text{new}} = \begin{cases} +1, & \text{if } w^T x_{\text{new}} + b \geq 0, \\ -1, & \text{if } w^T x_{\text{new}} + b < 0. \end{cases}$$

However, when the data are linearly separable, there are infinitely many possible separating hyperplanes. To improve generalisation to unseen data, we seek the **maximum-margin hyperplane** – the one that maximises the smallest distance (or *margin*) from any training point to the separating hyperplane.

(a) Show that the (signed) perpendicular distance from the hyperplane to a point x is given by

$$d(x) = \frac{w^T x + b}{\|w\|_2}.$$

[3%]

(b) Show that, with a suitable scaling, the margin is given by

$$r = \frac{1}{\|w\|_2}.$$

Explain your reasoning carefully.

[3%]

(c) Show that the maximum-margin hyperplane is the solution to the optimisation problem

$$\begin{aligned} \text{Maximise} \quad & \frac{1}{\|w\|_2} \\ \text{subject to} \quad & y_i(w^T x_i + b) \geq 1, \end{aligned}$$

where the optimisation is over (w, b).

[3%]

(d) Show that, instead of solving the problem in (c), we can solve:

$$\begin{aligned} \text{Minimise} \quad & \frac{1}{2}\|w\|_2^2 \\ \text{subject to} \quad & y_i(w^T x_i + b) \geq 1. \end{aligned}$$

Explain why this might be preferable.

[3%]

Question 2 continues on the following page

(e) Using the barrier method you implemented in Question 1, train your model on the training data provided in `part_e_training_data.csv` on Moodle. Row i corresponds to a single training data pair (x_i, y_i). The first two columns are the components of $x_i \in \mathbb{R}^2$. The third column is the associated label $y_i \in \{-1, 1\}$. If you were unable to complete Question 1, you may instead use the optimisation routines from a library of your choice (e.g. SciPy).

In your report, include a plot of the training data and the maximum-margin hyperplane you have found. What is the margin for the training data? Use your model to classify the points $(-2, 0.5)$ and $(3, 0)$. Discuss how each training point influences the maximum-margin hyperplane.

[10%]

(f) Not all data are linearly separable. However, some data are almost linearly separable, with the exception of a few outliers. To handle this, we can introduce slack variables $\xi_i \geq 0$, $i = 1, \ldots, m$, to allow some points to fall on the wrong side of the margin. This leads to the soft-margin problem:

$$\begin{aligned} \text{Minimise} \quad & \frac{1}{2}\|w\|_2^2 + c\|\xi\|_1 \\ \text{subject to} \quad & y_i(w^T x_i + b) \geq 1 - \xi_i, \\ & \xi \succeq 0, \end{aligned}$$

where $c > 0$ is a parameter and $\|\xi\|_1 \coloneqq \sum_{i=1}^m |\xi_i|$ is the ℓ_1 norm of ξ.

Explain why ξ appears in the objective function and what the parameter c does. How would changing the type of norm (i.e. to the ℓ_2 or ℓ_∞ norm) affect the optimal hyperplane?

[7%]

(g) Modify your solver from part (e) to solve the soft-margin problem in part (f). Train your model on an almost linearly separable dataset of your choice. You may use existing datasets, for example, from Kaggle (https://www.kaggle.com/) or `scikit-learn` (https://scikit-learn.org/stable/), or generate your own synthetic data. Discuss how well your model performs (in terms of e.g. margin size, number of misclassified points, etc.) as you change the parameter c. You may also wish to explore how partitioning data into train/validation/test[2] splits can allow you to choose c.

[14%]

(h) Does strong duality hold for the hard-margin problem in part (d)? Derive the dual problem for this problem.

[7%]

[2]You can train the model using the 'train' partition, assess its performance for a given parameter (e.g. c) using the 'validation' partition, and use the 'test' partition as a representation of the model's ability to classify unseen data.